



FLEET & SHORE PERSONNEL TRAINING

Risk Assessment and Hazard Identification Training

How to identify hazards before they cause harm, score and control the risk they present, and keep that assessment alive throughout the job.

ISM Code

Company SMS

STCW

What We Will Cover

01

Foundations of Risk Assessment

What risk assessment is and the ISM Code duty behind it

02

Hazard Identification Techniques

Finding hazards before they find you

03

The Risk Matrix

Scoring risk by likelihood x severity

04

Hierarchy of Controls

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Keeping the paperwork honest and current



SECTION 1

Foundations of Risk Assessment

01

Why the ISM Code requires every ship to systematically identify hazards before work begins, not after something goes wrong.

The ISM Code Requirement

Risk assessment is not a paperwork exercise — it is a core element of the Safety Management System required by the ISM Code.

- ISM Code paragraph 1.2.2.2 requires the Company to assess all identified risks to its ships, personnel and the environment, and establish appropriate safeguards.
- Element 7 (Shipboard Operations) requires procedures for the safe conduct of operations, including risk assessment for tasks that are non-routine or unfamiliar.
- Element 8 (Emergency Preparedness) is itself built on risk assessment: identifying credible shipboard emergencies before drilling for them.
- STCW competence tables require officers and ratings to demonstrate the ability to identify hazards and apply safe working practices in their own discipline.
- The Company SMS translates these requirements into a practical form: a risk assessment (RA) must be completed and reviewed before non-routine or higher-risk work starts.



Reasonably Practicable

Controls must reduce risk 'as low as reasonably practicable' — weighing the cost, time and effort of a control against the reduction in risk it actually delivers.

THE CORE PROCESS

Three Steps, One Process

Every risk assessment, however simple or complex, follows the same three-step logic.

1

Identify — break the task into steps and find the hazards present in each one

2

Assess — score each hazard's likelihood and severity to determine the risk level

3

Control — apply the hierarchy of controls, record the assessment, and monitor its effectiveness

KEY DEFINITIONS

Terms Every Crew Member Should Know

A shared vocabulary keeps risk assessments consistent across the fleet.



Hazard

Anything with the potential to cause harm — to people, the ship, cargo or the environment.



Risk

The combination of the likelihood that a hazard will cause harm and the severity of that harm.



Likelihood

How probable it is that the hazard will actually result in an incident, given current controls.



Severity

How serious the outcome would be if the hazard did result in an incident.



SECTION 2

Hazard Identification Techniques

02

Systematic ways to surface hazards that a quick glance around the workplace will miss.

METHODS

Finding Hazards Before They Find You

No single method catches everything — effective hazard identification combines several sources.

- Break the job into individual steps and walk through each one, asking what could go wrong at that specific point.
- Physically inspect the workplace, equipment and access routes immediately before the task, not from memory of how it 'usually' looks.
- Review near-miss and incident reports from similar jobs, on this ship and across the fleet, for hazards others have already encountered.
- Consult the crew members who will actually perform the task — their hands-on experience often reveals hazards a procedure alone would miss.
- Check equipment manuals, manufacturer warnings and the Company SMS for hazards already documented for that machine or operation.



Look, Then Look Again

Conditions change during a job — weather, lighting, adjacent work, fatigue. Hazard identification is not a one-off step; it continues for as long as the task runs.

STRUCTURED APPROACHES

Four Practical Techniques

These techniques are used together, not as alternatives to one another.



Job Hazard Analysis (JHA)

Breaks a task into sequential steps and identifies the specific hazard and control for each step.



Workplace Walk-Through

A physical inspection of the work area, equipment and access immediately before work starts.



Near-Miss & Incident Review

Mines the Company's own reporting data for hazards that have already caused harm or nearly did.



Crew Consultation

Draws on the direct experience of the people who will do the job, including new joiners' fresh perspective.

CATEGORIES TO CHECK

Common Hazard Categories Aboard Ship

Working through each category in turn reduces the chance of overlooking a hazard type entirely.

M

Mechanical — moving parts, pinch points, falling or shifting loads

E

Electrical — live circuits, damaged cables, inadequate isolation

C

Chemical — fuels, cleaning agents, cargo residues, fumes

H

Human Factors — fatigue, communication, unfamiliarity, workload



SECTION 3

The Risk Matrix

Turning a hazard into a number: how likelihood and severity combine to set the risk level and the response it demands.

03

THE METHOD

Likelihood x Severity

The risk matrix gives every hazard a consistent, comparable score.

- Each hazard is scored on two independent scales: how likely it is to occur, and how severe the outcome would be if it did.
- Most Company SMS matrices use a 5x5 grid, with likelihood and severity each scored from 1 (rare / negligible) to 5 (almost certain / catastrophic).
- The risk score is the product of the two ratings — a likely event with minor consequences can score the same as a rare event with severe consequences.
- Scoring should reflect the risk with existing controls already in place, not the theoretical worst case with no controls at all.
- Where the team disagrees on a score, the higher (more cautious) rating should normally be used until the disagreement is resolved.



Consistency Matters

The exact numbers matter less than applying them consistently — a 4 should mean the same thing in the engine room as it does on deck.

THE RATING SCALES

Likelihood and Severity Levels

A typical 5-point scale used across the Company's risk assessments.

Score	Likelihood	Severity
1	Rare — practically never expected	Negligible — no injury, no lost time
2	Unlikely — could occur, but not expected	Minor — first aid only
3	Possible — could occur at some point	Moderate — lost-time injury
4	Likely — expected to occur eventually	Major — serious injury or damage
5	Almost Certain — expected in most circumstances	Catastrophic — fatality, loss of ship

RISK BANDS

From Score to Required Action

The resulting score is grouped into bands, each with a defined minimum response.

Low

Existing routine controls are sufficient; proceed with standard precautions

Medium

Additional controls needed; review by supervisor or Officer of the Watch

High

Senior officer / Master approval required before work proceeds

Extreme

Stop work; escalate to the Master and Company DPA before any further action



SECTION 4

Hierarchy of Controls

Not all controls are equally effective — the hierarchy ranks them from most to least reliable.

04

THE PRINCIPLE

From Elimination to PPE

The hierarchy of controls ranks control options by how reliably they reduce risk, not by how easy they are to apply.

- Controls higher in the hierarchy remove or reduce the hazard itself; controls lower down rely on people behaving correctly, every time, under all conditions.
- A competent risk assessment considers each level in order, starting at the top, before settling for a lower-ranked control.
- In practice, most real-world jobs end up using a combination of controls from several levels at once, not a single measure alone.
- Personal protective equipment (PPE) is the last line of defence — it protects the individual only after all higher controls have already been applied.
- Selecting PPE as the sole control for a hazard that could instead be eliminated or engineered out is a common and serious risk assessment failure.



PPE Is Not a Substitute

PPE only works if it is worn correctly, fits properly, and is maintained — a single point of failure that higher-level controls do not share.

THE FIVE LEVELS

The Hierarchy of Controls

Ranked from the most effective control at the top to the least effective at the bottom.



1. Elimination

Remove the hazard entirely — for example, doing a task from the deck instead of aloft.



2. Substitution

Replace the hazard with something less dangerous, such as a safer cleaning chemical.



3. Engineering Controls

Isolate people from the hazard with guarding, ventilation, interlocks or barriers.



4. Administrative Controls

Change the way people work: procedures, permits, training, signage and supervision.



5. PPE

Protect the individual as the final layer once every higher-level control has been applied.

Choosing Controls Correctly

The order in which controls are considered matters as much as which controls are chosen.



Do

- Start at the top of the hierarchy and work down, one level at a time
- Combine controls from more than one level where practicable
- Record why a higher-level control was ruled out, not just which one was chosen
- Review whether a higher-level control has since become practicable



Don't

- Don't jump straight to PPE because it is the fastest option
- Don't assume a control that worked once will always remain effective
- Don't treat administrative controls (signs, procedures) as a substitute for a physical barrier
- Don't close out the assessment without checking the control was actually implemented



SECTION 5

Toolbox Talks

A short briefing, held at the worksite, that turns a written risk assessment into a shared understanding.

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PURPOSE

Briefing the Crew Before the Job Starts

A toolbox talk is the point where the risk assessment on paper becomes a shared plan in the team's heads.

- Held immediately before the task, at or near the worksite, with everyone who will actually take part in the job present.
- Walks the team through the task steps, the hazards identified, the controls in place, and each person's individual role.
- Gives every team member a chance to raise a hazard or condition the written assessment did not anticipate.
- Confirms that PPE, tools and equipment required by the risk assessment are actually on hand and in serviceable condition.
- Is repeated whenever the task is interrupted and resumed, the team changes, or conditions change significantly.



Stop-Work Authority

Every crew member, regardless of rank, has the authority and the duty to stop a job the moment a new or unassessed hazard appears.

CONTENT

What a Good Toolbox Talk Covers

A consistent structure keeps toolbox talks short, useful and easy to repeat.



Task & RA Review

A quick recap of the job's scope and the risk assessment's key findings.



Hazards & Controls

The specific hazards expected today and the controls the team will actually use.



Roles & Communication

Who does what, and how the team will communicate, including emergency signals.



Stop-Work Authority

A reminder that anyone can halt the job if something unexpected appears.

Conducting an Effective Toolbox Talk

How the talk is run determines whether it changes behaviour or is treated as a formality.



Do

- Keep it short, specific to today's job, and held at the actual worksite
- Ask open questions and invite every participant to speak, not just listen
- Confirm required PPE and equipment are physically present before starting
- Record attendance and any hazards raised that were not in the written RA



Don't

- Don't read the risk assessment aloud word for word as a substitute for discussion
- Don't hold the talk somewhere the crew cannot see the actual work area
- Don't skip the talk because 'we've done this job a hundred times before'
- Don't proceed if a participant raises a hazard that has not yet been addressed



SECTION 6

Permit-to-Work Systems

Formal, signed authorisation for the highest-risk jobs aboard — a second layer of control beyond the toolbox talk.

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PURPOSE

When a Permit Is Required

A permit-to-work (PTW) is a formal, documented check that a job's specific precautions are in place before it starts.

- Required for defined categories of higher-risk work: enclosed space entry, hot work, work aloft or over the side, and electrical isolation, among others.
- Names the specific job, its time window, the precautions required and the individuals authorised to carry it out.
- Is issued by a responsible officer only after the required checks — gas testing, isolation, area preparation — have actually been completed.
- Remains valid only for its stated conditions and time period; any change in scope or conditions requires the permit to be reissued.
- Must be formally closed out on completion, confirming the area has been made safe and equipment restored to normal condition.



Enclosed Space Entry

Enclosed space entry permits require atmosphere testing before entry and continuous monitoring throughout, with a dedicated standby person at the entrance at all times.

CATEGORIES

Common Permit-to-Work Categories

Each category carries its own specific pre-checks and precautions.

Permit Type	Typical Hazard	Key Precaution
Enclosed Space Entry	Oxygen deficiency, toxic or flammable gas	Atmosphere testing and continuous monitoring
Hot Work	Fire, explosion from sparks or heat	Fire watch, area clearance, extinguisher on hand
Working Aloft / Overside	Fall from height, man overboard	Harness, lifeline, standby person, calm weather
Electrical Isolation	Electric shock, unexpected re-energisation	Lock-out / tag-out and voltage verification

ROLES

Who Does What in the Permit System

A permit involves more than the person signing it — clear roles keep the system reliable.



Permit Issuer

A responsible officer who verifies precautions are in place and authorises the work in writing.



Performing Authority

The person-in-charge on site, responsible for the job being carried out as permitted.



Standby Person

Stationed at the entry point or work area to raise the alarm and assist if something goes wrong.



Master / Responsible Officer

Holds overall authority to suspend or cancel a permit at any time if conditions change.



SECTION 7

Recording & Reviewing Assessments

07

A risk assessment is a living document — its value depends on being recorded properly and revisited often enough.

Recording the Assessment

A well-recorded assessment allows anyone to understand the job's risks without having to rediscover them.

- Record the task, the hazards identified, the likelihood and severity scores, and the controls selected for each hazard.
- Name the people involved in the assessment and the date it was carried out, together with the officer who approved it.
- Keep the record in a format the whole crew can access, following the format and retention period set by the Company SMS.
- A generic risk assessment can be adapted for a specific job, but the adaptation itself must be recorded, not just assumed.
- Illegible, incomplete or copy-pasted assessments are treated by auditors and PSC inspectors as evidence of a weak safety culture.



Sign It Personally

Every participant should sign or otherwise confirm they understood the assessment — a shared signature list is not the same as individual understanding.

REVIEW TRIGGERS

When to Review a Risk Assessment

A risk assessment should never simply be filed and forgotten.

Before

A non-routine, dissimilar or unfamiliar version of a previously assessed job

After

Any incident, near miss or unexpected event during or after the job

Periodically

At the interval defined in the Company SMS, even if nothing has gone wrong

From Record to Continuous Improvement

Recorded assessments feed the wider safety management system, not just the individual job file.



Retention & Access

Completed assessments are retained and remain accessible for audit, investigation and future reference.



Internal Audit & Verification

Internal audits and Master's reviews check that assessments reflect what actually happens on board.



Lessons Learned

Findings from incidents and near misses are fed back into generic risk assessments fleet-wide.



Crew Feedback

Crew input on assessments that do not match real working conditions is actively sought and acted on.



COURSE COMPLETE

A risk assessed on paper only protects the crew once it is understood, applied and kept alive on deck.

Company DPA — 24/7 emergency line

Nearest RCC / MRCC — via VHF Ch.16 / DSC

Fleet Safety Department — mng@gloryship.com.tr